

AND GATE

The screenshot shows the VS Code interface with the Verilog code for an AND gate using behavioural modeling. The Explorer panel on the left shows the project structure with files like `and_gate.v`, `and_gate_tb.v`, and `and_gate.vcd`. The main editor displays the code for `and_gate.v`, which includes a module definition with inputs `a` and `b`, and output `c`. The code uses an `always` block with a sensitivity list `@(a,b)` to model the AND gate's behavior. The terminal at the bottom shows the command `iverilog -o and and_gate.v and_gate_tb.v` and the output `vvp and`.

```
1 //Behavioural Modeling
2 module and_gate(
3     input a,
4     input b,
5     output reg c
6 );
7
8 always @(a,b) //Sensitivity list
9 begin
10     if(a==0 && b==0)
11         c<=0;
12     else if(a==0 && b==1)
13         c<=0;
14     else if(a==1 && b==0)
15         c<=0;
16     else if(a==1 && b==1)
17         c<=1;
18 end
19
20 endmodule
```

BEHAVIOURAL MODELING

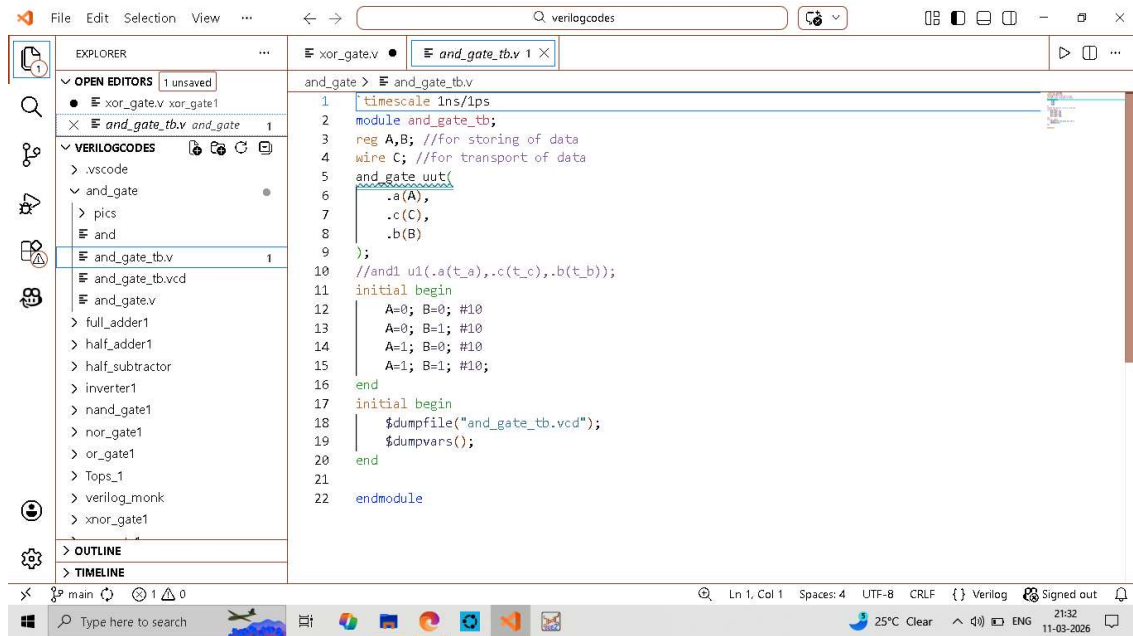
The screenshot shows the VS Code interface with the Verilog code for an AND gate using gate level modeling. The Explorer panel on the left shows the project structure with files like `and_gate.v`, `and_gate_tb.v`, and `and_gate.vcd`. The main editor displays the code for `and_gate.v`, which includes a module definition with inputs `a` and `b`, and output `c`. The code uses a simple `and` gate primitive to model the AND gate's behavior. The terminal at the bottom shows the command `iverilog -o and and_gate.v and_gate_tb.v` and the output `vvp and`.

```
21 //Gate Level Modeling
22 module and_gate(
23     input a,
24     input b,
25     output c
26 );
27     and(c,a,b);
28 endmodule
29
30 //Data Flow Modeling
31 module and_gate(
32     input a,
33     input b,
34     output c
35 );
36
37
38 assign c=(a&b);
39 endmodule
```

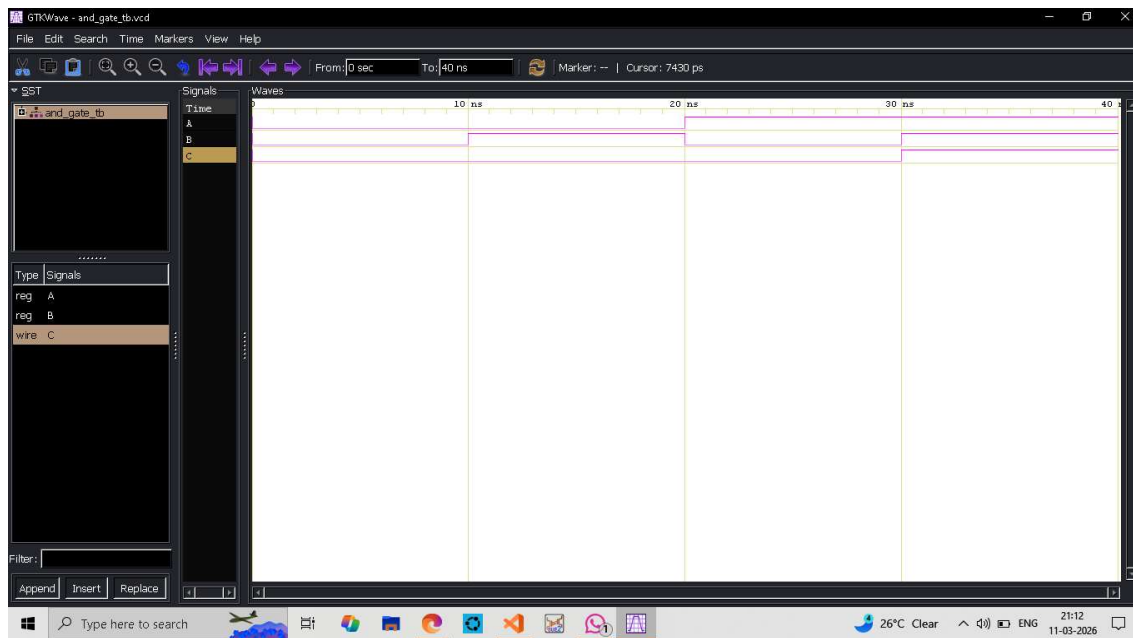
GATE LEVEL MODELING

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DATAFLOW MODELING



TESTBENCH



WAVEFORM

OR GATE

The screenshot shows the Verilog code for an OR gate using behavioral modeling. The code is as follows:

```
1 //Behavioural Modeling
2 module or_gate(
3     input a,
4     input b,
5     output reg c
6 );
7
8 always @(a,b) //Sensitivity list
9 begin
10     if(a==0 && b==0)
11         c<=0;
12     else
13         c<=1;
14 end
15
16 endmodule
```

The terminal window shows the following commands and output:

```
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> iverilog -o or or_gate.v or_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> vvp or
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> gtkwave or_gate.vcd
```

BEHAVIOURAL MODELING

The screenshot shows the Verilog code for an OR gate, illustrating both gate level and data flow modeling. The code is as follows:

```
18 //Gate Level Modeling
19 module or_gate(
20     input a,
21     input b,
22     output c
23 );
24
25 or(c,a,b);
26
27 endmodule
28
29 //Data Flow Modeling
30 module or_gate(
31     input a,
32     input b,
33     output c
34 );
35
36 assign c=(a|b);
37 endmodule
```

The terminal window shows the same commands and output as the previous screenshot:

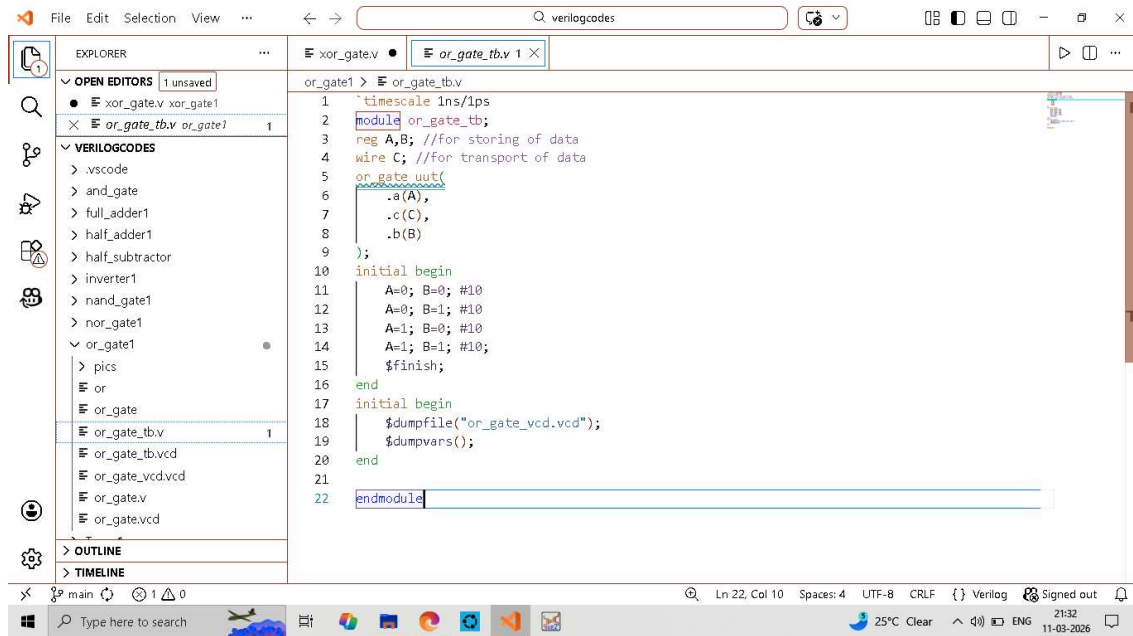
```
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> iverilog -o or or_gate.v or_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> vvp or
PS C:\Users\HP\Documents\VS_code\verilogcodes\or_gate1> gtkwave or_gate.vcd
```

GATE LEVEL MODELING

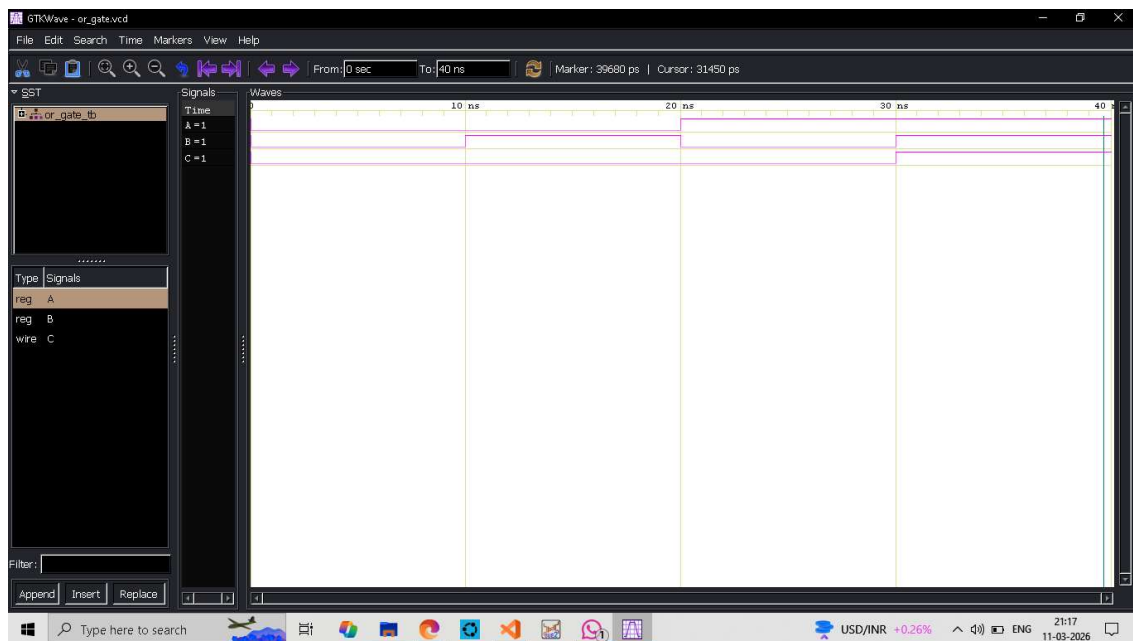
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DATAFLOW MODELING

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TESTBENCH



WAVEFORM

NAND GATE

The screenshot shows the Verilog code for a NAND gate using behavioural modeling. The code is as follows:

```
1 //Behavioural Modeling
2 module nand_gate(
3     input a,
4     input b,
5     output reg c
6 );
7
8 always @(a,b) //Sensitivity list
9 begin
10     if(a==1 && b==1)
11         c<=0;
12     else
13         c<=1;
14 end
15
16 endmodule
```

The terminal window shows the following commands and output:

```
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> iverilog -o nand nand_gate.v nand_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> vvp nand
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> gtkwave nand_gate.vcd
```

BEHAVIOURAL MODELING

The screenshot shows the Verilog code for a NAND gate using gate level and data flow modeling. The code is as follows:

```
18 //Gate Level Modeling
19 module nand_gate(
20     input a,
21     input b,
22     output c
23 );
24 wire c_1;
25 and(c_1,a,b);
26 not(c,c_1);
27
28 endmodule
29 //Data Flow Modeling
30 module nand_gate(
31     input a,
32     input b,
33     output c
34 );
35
36 assign c=~(a&b);
37 endmodule
```

The terminal window shows the following commands and output:

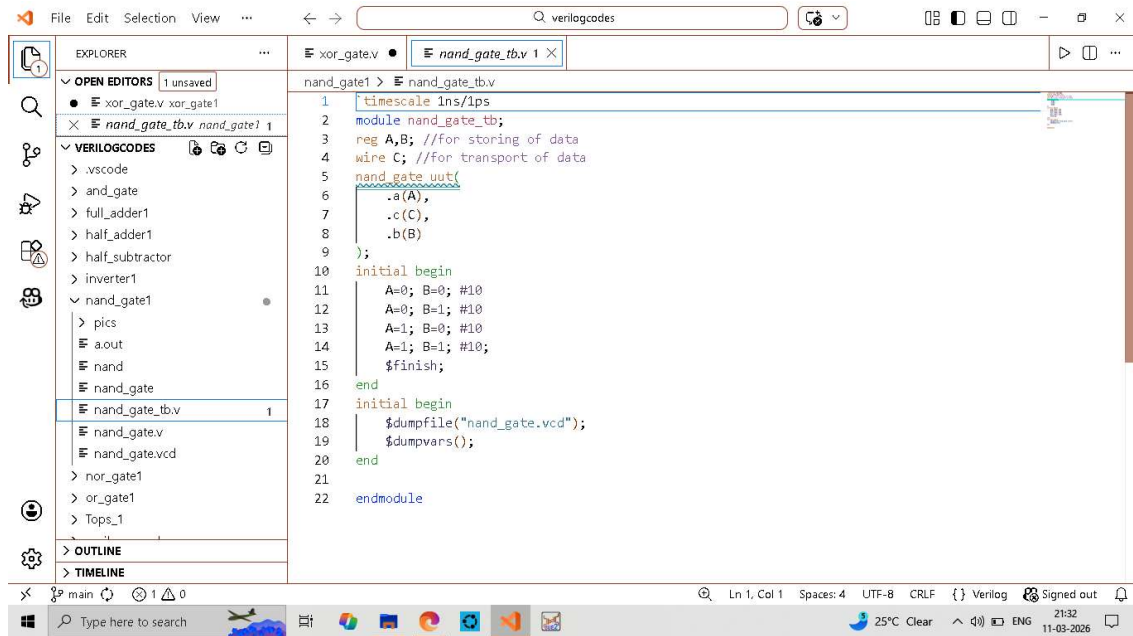
```
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> iverilog -o nand nand_gate.v nand_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> vvp nand
PS C:\Users\HP\Documents\VS_code\verilogcodes\nand_gate1> gtkwave nand_gate.vcd
```

GATE LEVEL MODELING

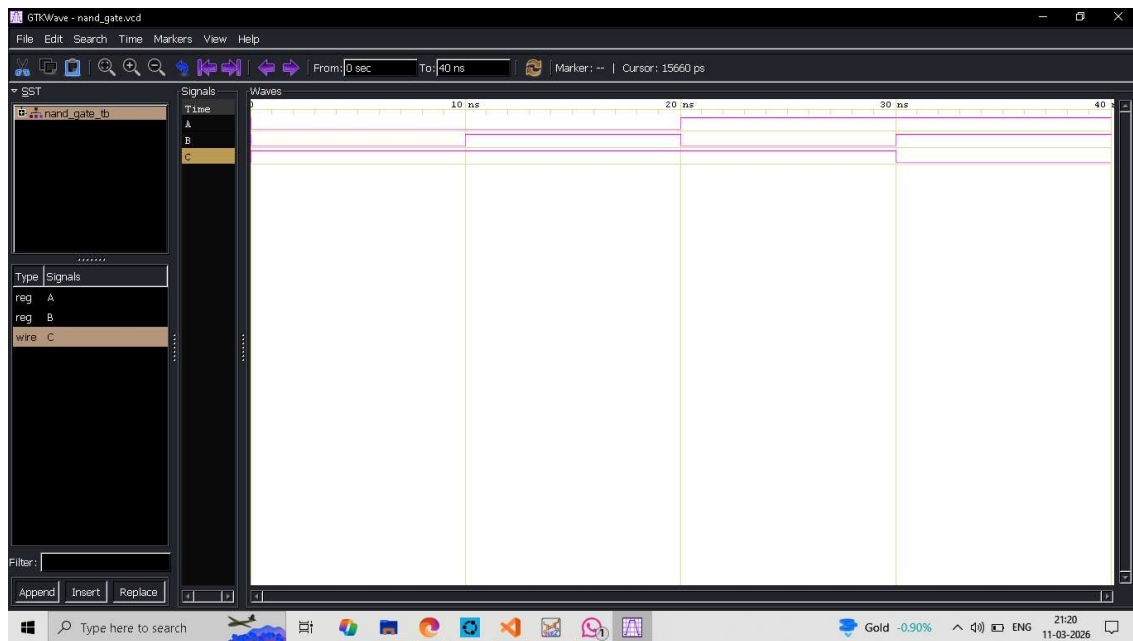
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WAVEFORM

NOR GATE

The screenshot shows the VS Code interface with the 'nor_gate.v' file open. The code implements a NOR gate using behavioural modeling. The Explorer panel on the left shows the project structure with files like 'nor_gate1_tb.v', 'and_gate.v', 'or_gate.v', 'nand_gate.v', 'nor_gate.v', and 'xnor_gate1.v'. The Verilog code in the main editor is as follows:

```
1 //Behavioural Modeling
2 module nor_gate(
3     input a,
4     input b,
5     output reg c
6 );
7
8 always @(a,b) //Sensitivity list
9 begin
10     if(a==0 && b==0)
11         c<=1;
12     else
13         c<=0;
14 end
15
16 endmodule
```

The terminal at the bottom shows the execution of the following commands:

```
PS C:\Users\HP\Documents\VS_code\verilogcodes\nor_gate1> iverilog -o nor nor_gate.v nor_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\nor_gate1> vvp nor
VCD info: dumpfile nor_gate.vcd opened for output.
nor_gate_tb.v:15: $finish called at 40000 (1ps)
PS C:\Users\HP\Documents\VS_code\verilogcodes\nor_gate1> gtkwave nor_gate.vcd
```

BEHAVIOURAL MODELING

The screenshot shows the VS Code interface with the 'nor_gate.v' file open. The code implements a NOR gate using gate level and data flow modeling. The Explorer panel on the left shows the project structure with files like 'nor_gate1_tb.v', 'and_gate.v', 'or_gate.v', 'nand_gate.v', 'nor_gate.v', and 'xnor_gate1.v'. The Verilog code in the main editor is as follows:

```
18 //Gate Level Modeling
19 module nor_gate(
20     input a,
21     input b,
22     output c
23 );
24 wire c_1;
25 or(c_1,a,b);
26 not(c,c_1);
27 endmodule
28
29 //Data Flow Modeling
30 module nor_gate(
31     input a,
32     input b,
33     output c
34 );
35
36 assign c=~(a|b);
37 endmodule
```

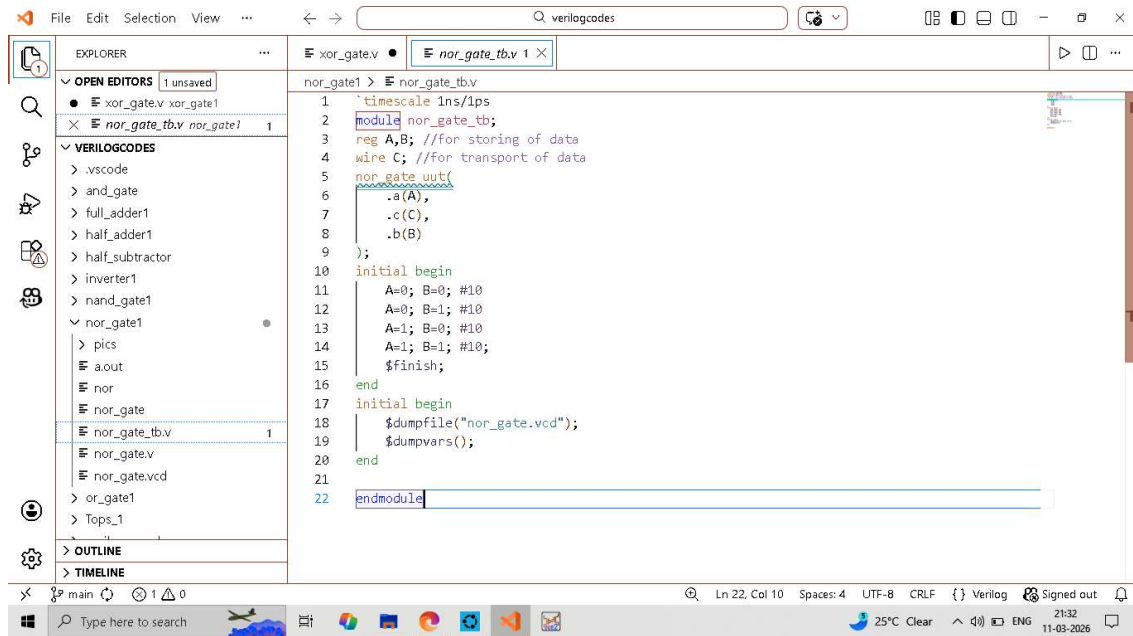
The terminal at the bottom shows the execution of the following commands:

```
VCD info: dumpfile nor_gate.vcd opened for output.
nor_gate_tb.v:15: $finish called at 40000 (1ps)
PS C:\Users\HP\Documents\VS_code\verilogcodes\nor_gate1> gtkwave nor_gate.vcd
```

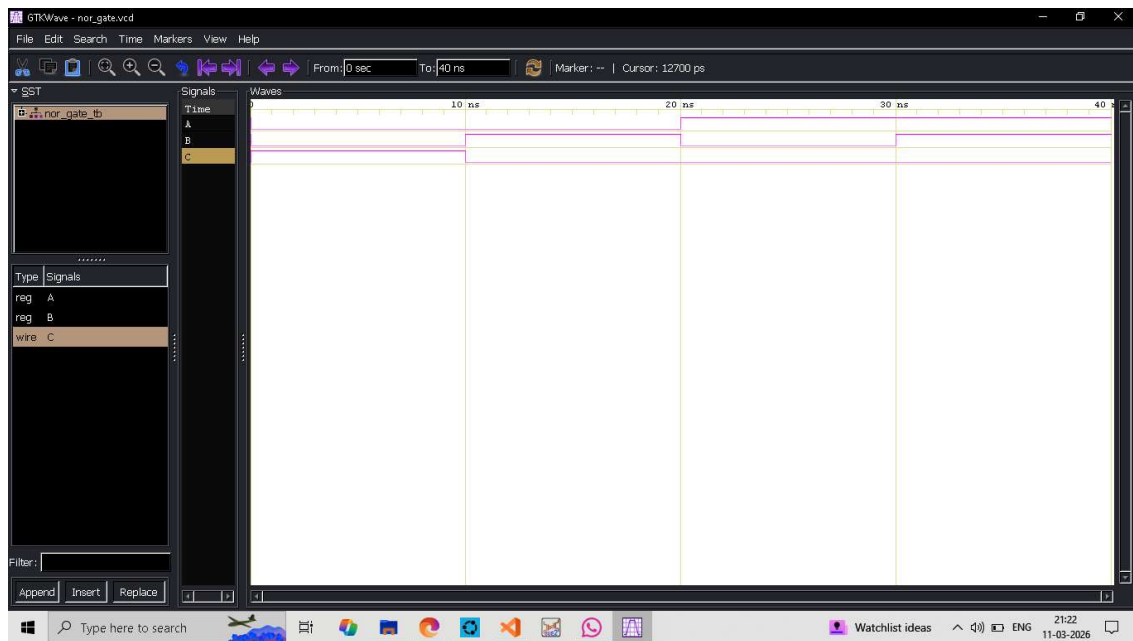
GATE LEVEL MODELING

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DATAFLOW MODELING



TESTBENCH



WAVEFORM

XNOR GATE

The screenshot shows the Verilog code for an XNOR gate using behavioural modeling. The code is as follows:

```
1 //Behavioural Modeling
2 module xnor_gate1(
3     input a,
4     input b,
5     output reg c
6 );
7
8 always @(a,b) //Sensitivity list
9 begin
10     if(a==0 && b==0)
11         c<=1;
12     else if(a==0 && b==1)
13         c<=0;
14     else if(a==1 && b==0)
15         c<=0;
16     else
17         c<=1;
18 end
19
20 endmodule
```

The Explorer panel on the left shows the project structure with files like `xnor_gate1_tb.v`, `xnor_gate1.v`, and `xnor_gate.vcd`. The Terminal panel at the bottom shows the command `iverilog -o xnor xnor_gate1.v xnor_gate1_tb.v` and the output `VCD info: dumpfile xnor_gate.vcd opened for output.`

BEHAVIOURAL MODELING

The screenshot shows the Verilog code for an XNOR gate using gate level and data flow modeling. The code is as follows:

```
22 //Gate Level Modeling
23 module xnor_gate1(
24     input a,
25     input b,
26     output c
27 );
28     xor(c,a,b);
29 endmodule
30
31 //Data Flow Modeling
32 module xnor_gate1(
33     input a,
34     input b,
35     output c
36 );
37
38     assign c=(a^b);
39 endmodule
40
```

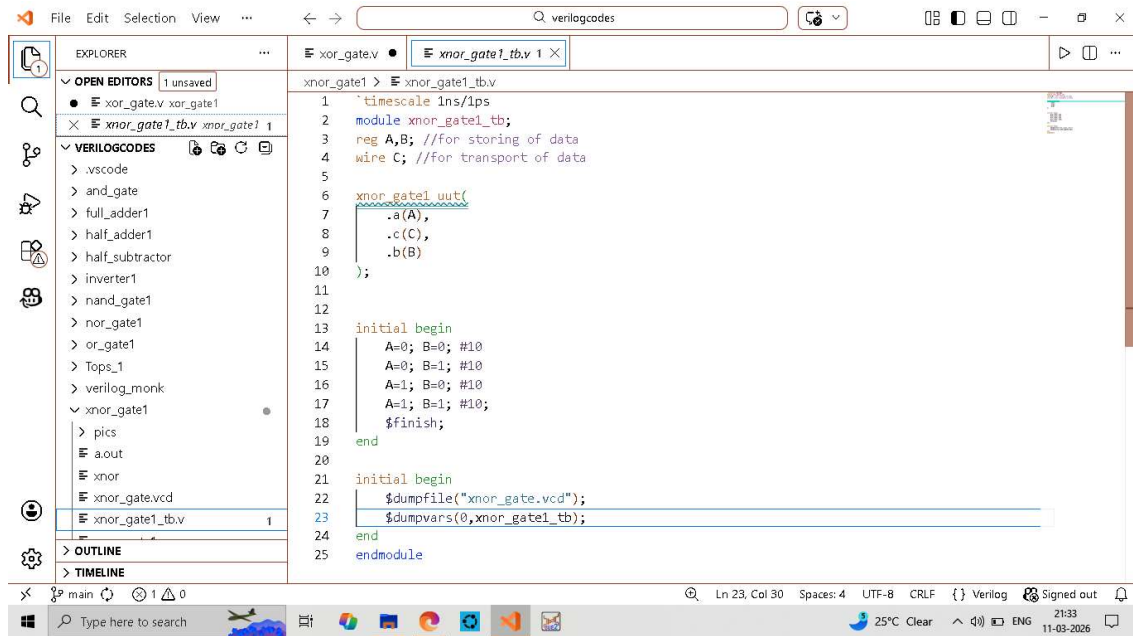
The Explorer panel on the left shows the project structure with files like `xnor_gate1_tb.v`, `xnor_gate1.v`, and `xnor_gate.vcd`. The Terminal panel at the bottom shows the command `gtkwave xnor_gate.vcd`.

GATE LEVEL MODELING

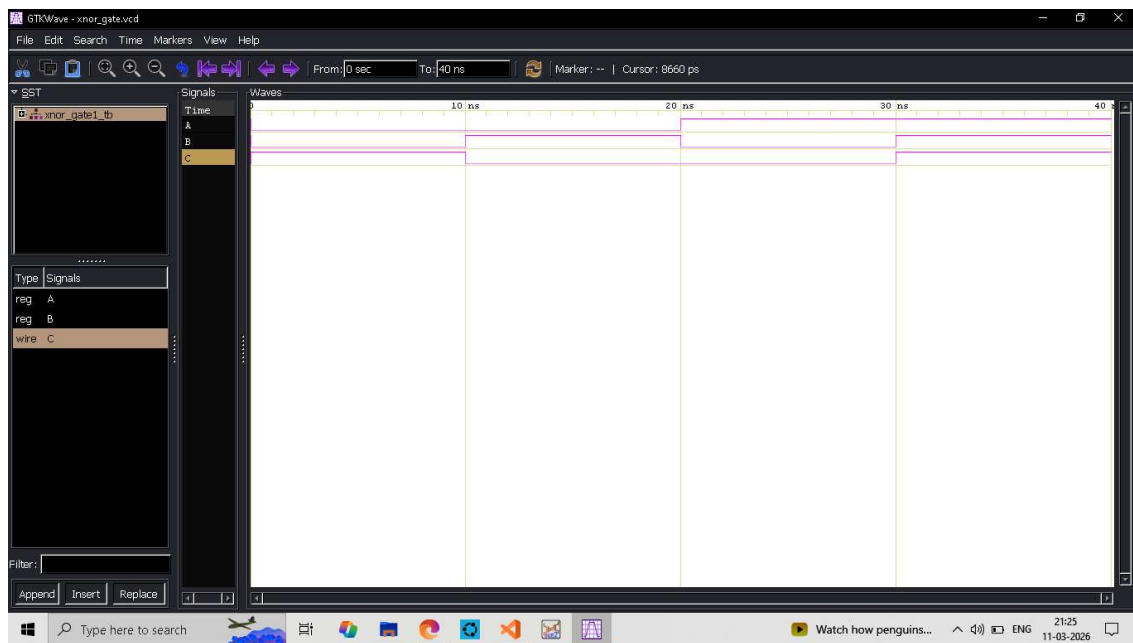
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DATAFLOW MODELING

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WAVEFORM

XOR GATE

The screenshot shows the Verilog code for an XOR gate in two styles: Gate Level Modeling and Data Flow Modeling. The code is written in a text editor with a sidebar showing the project structure. The terminal at the bottom shows the commands used to compile and simulate the code.

```
//Gate Level Modeling
module xor_gate(
    input a,
    input b,
    output c
);
    xor(c,a,b);
endmodule

//Data Flow Modeling
module xor_gate(
    input a,
    input b,
    output c
);
    assign c=(a^b);
endmodule
```

Terminal output:

```
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> iverilog -o xor xor_gate.v xor_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> vvp xor
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> gtkwave xor_gate.vcd
```

BEHAVIOURAL MODELING

The screenshot shows the Verilog code for an XOR gate using Behavioral Modeling. The code is written in a text editor with a sidebar showing the project structure. The terminal at the bottom shows the commands used to compile and simulate the code.

```
//Behavioural Modeling
module xor_gate(
    input a,
    input b,
    output reg c
);
    always @(a,b) //Sensitivity list
    begin
        if(a==0 && b==0)
            c<=0;
        else if(a==0 && b==1)
            c<=1;
        else if(a==1 && b==0)
            c<=1;
        else
            c<=0;
    end
endmodule
```

Terminal output:

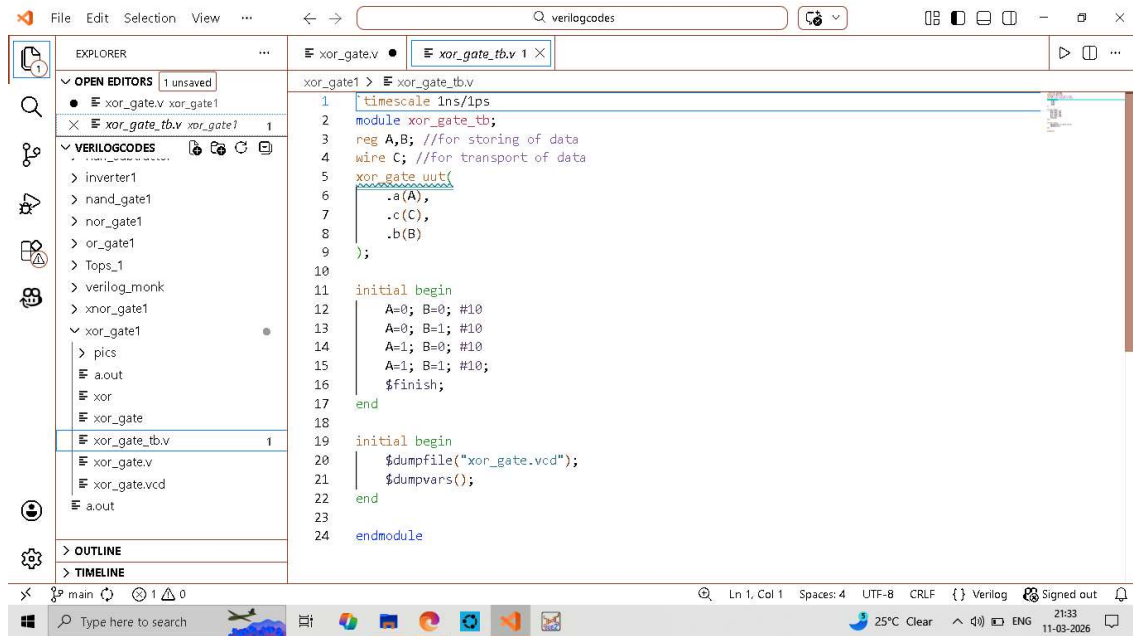
```
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> iverilog -o xor xor_gate.v xor_gate_tb.v
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> vvp xor
PS C:\Users\HP\Documents\VS_code\verilogcodes\xor_gate1> gtkwave xor_gate.vcd
```

GATE LEVEL MODELING

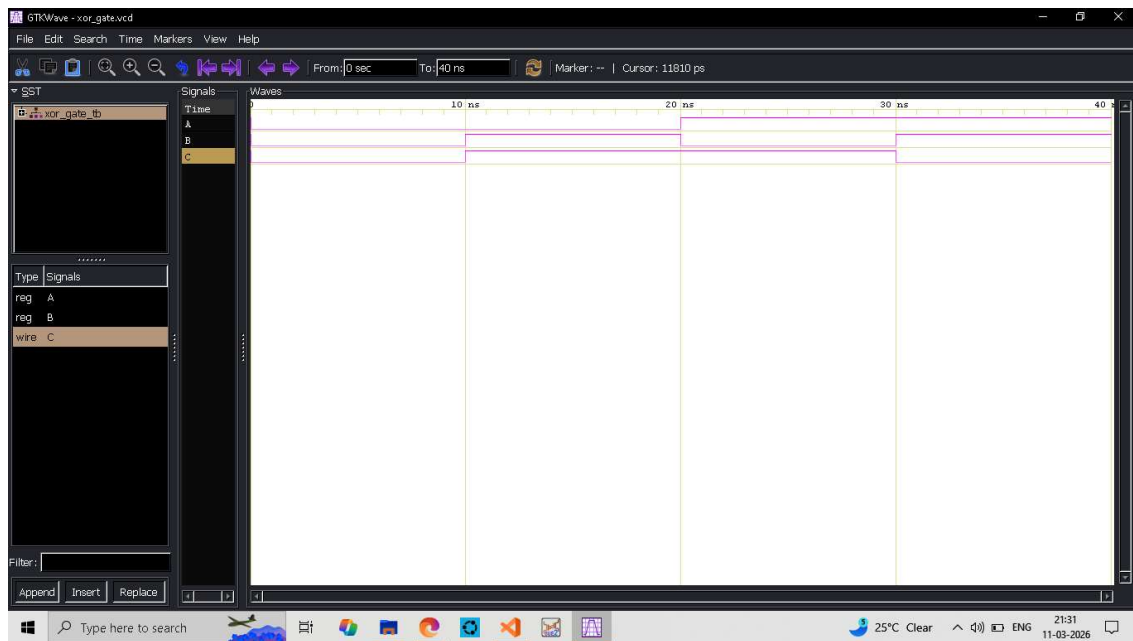
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WAVEFORM

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